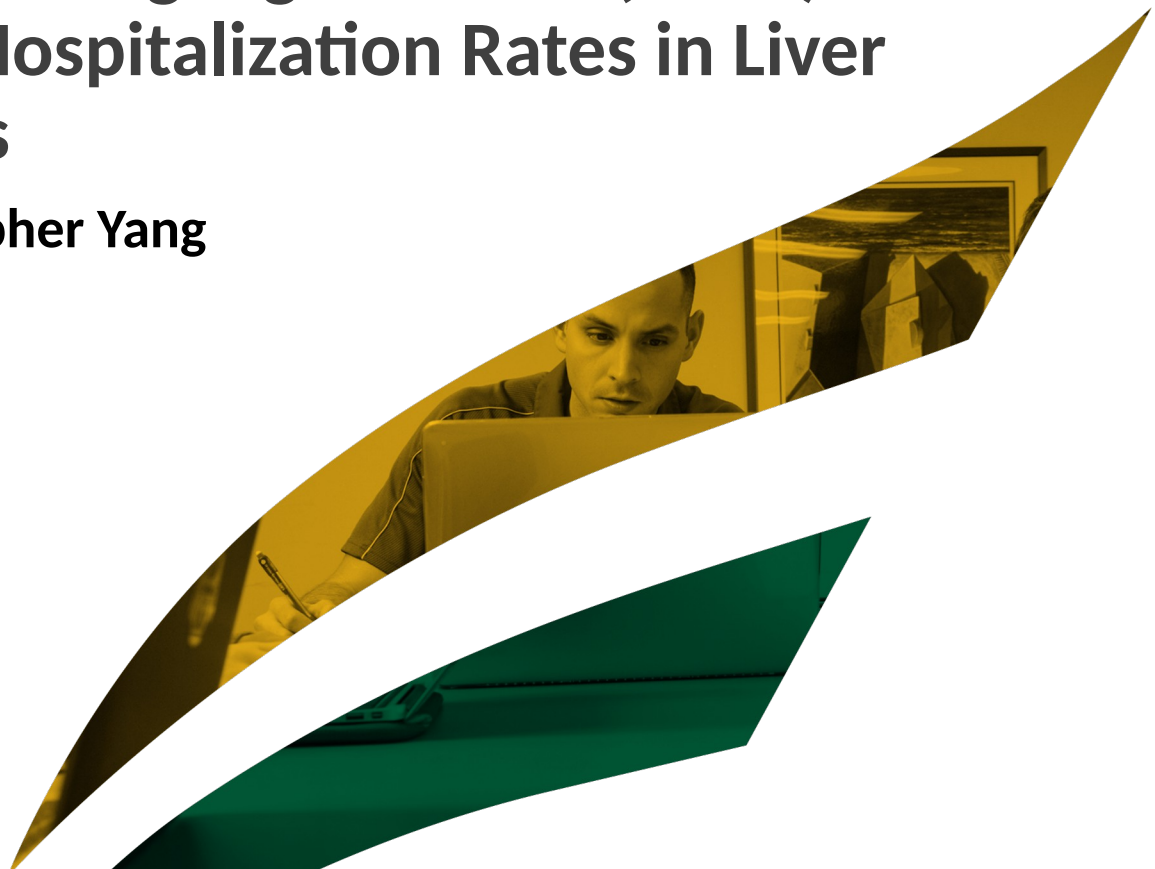
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Project 2: Using Large Language Models (LLMs) to Predict One-Year Hospitalization Rates in Liver Transplant Recipients

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Background

- LLM = Large Language Models
  - Why?
    - vs. ML, for predicting hospitalization rates
- Intro to liver transplants
- Prompting strategies:
  - Zero-shot
  - Few-shot
  - Chain-of Thought

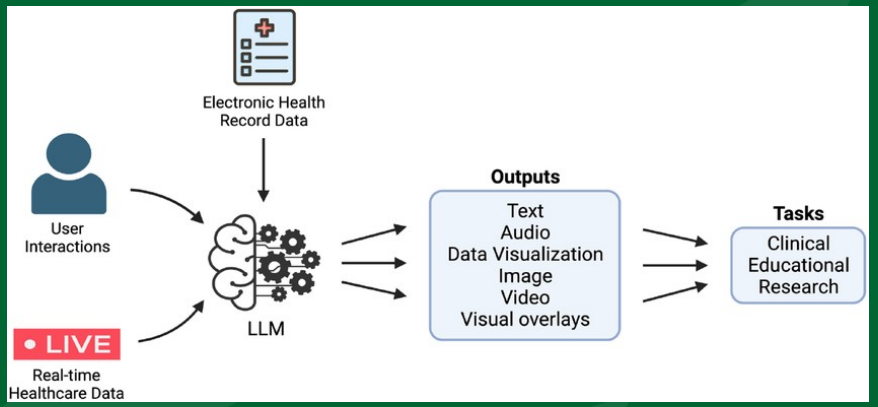


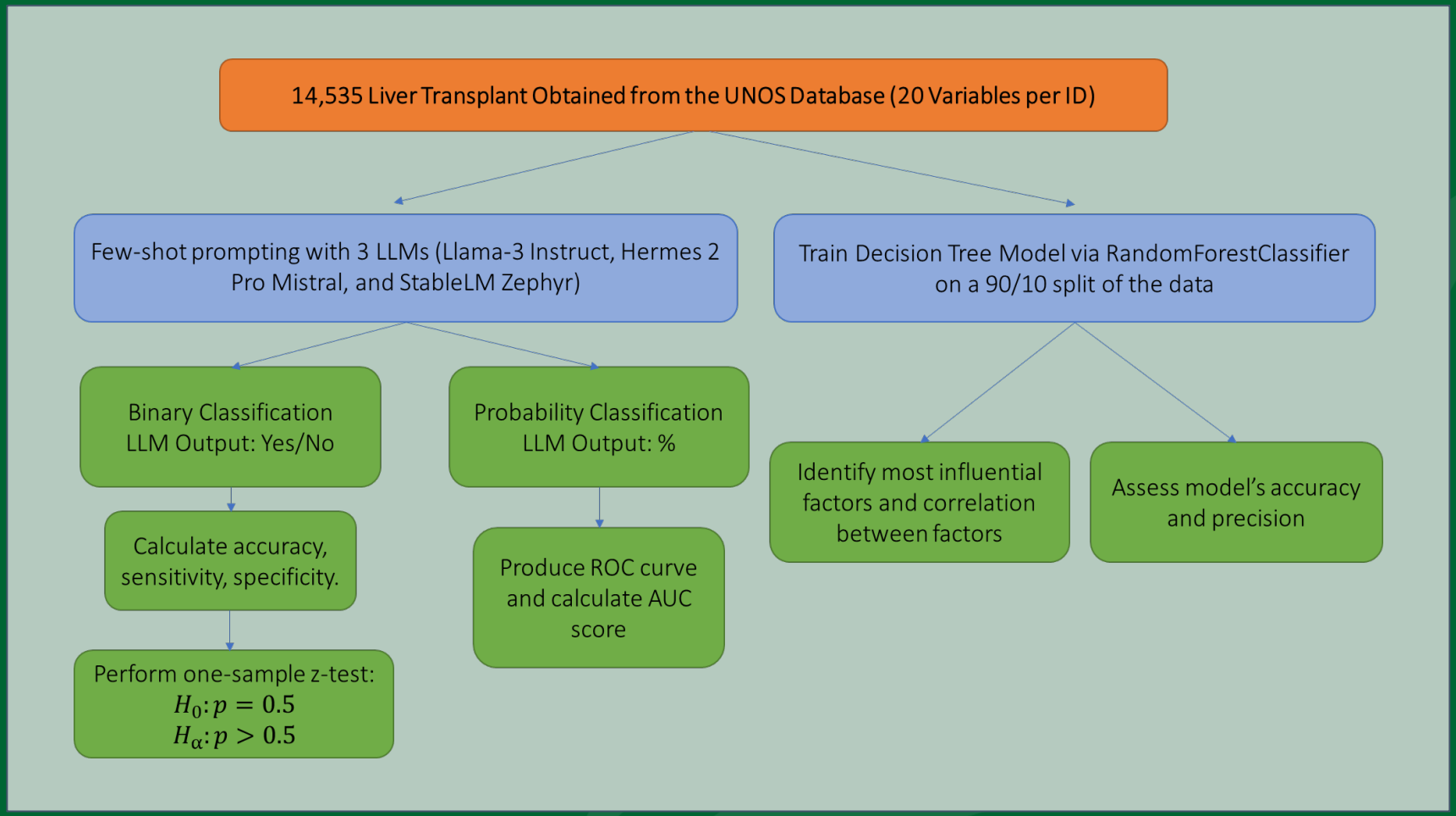
Figure 1. LLM Framework (Shahab et al., 2024)

Study Objective

Investigate the Efficacy of LLMs in Predicting One-Year Hospitalization Rates of Liver Transplant Recipients

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Methods



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Excerpt of Prompting

AGE	male_pt	black_pt	white_pt	hisp_pt	asian_pt	other_pt	diab	retx	FINAL_MELD_P	DAYSWAIT_CHR
63	0	0	0	1	0	0	0	0	11	101
40	0	0	0	1	0	0	1	0	38	3
19	0	1	0	0	0	0	0	1	33	19

```
pt_gender_code = row['male_pt']
if pt_gender_code == 1:
    pt_gender = "male"
elif pt_gender_code == 0:
    pt_gender = "female"
```

In this example, the patient's age (25) is relatively young. You should factor this reduced risk into your prediction.
The also patient has a moderately high FINAL_MELD_PELD_LAB_SCORE (19), indicating some risk but not extreme...

A {pt_age}-year-old {pt_race} {pt_gender} liver transplant recipient had a BMI of {pt_BMI} and a MELD score of {pt_meld} at the time of the transplant...
Using your knowledge of historical clinical data to inform your prediction, predict whether this patient will remain in hospital after one year.

Binary Classification

* When we used all 20 variables in the prompt on Hermes 2 Pro Mistral, all of the results were 1.

Model	Accuracy Rate	Sensitivity	Specificity	P-value
Llama 3 Instruct	0.588	0.829	0.222	0.073
Hermes 2 Pro Mistral	0.596	0.700	0.462	0.034
Stablelm Zephyr	0.472	0.240	0.769	0.298

Table 1. Results from using 11 variables (n = 90):

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Probability Classification

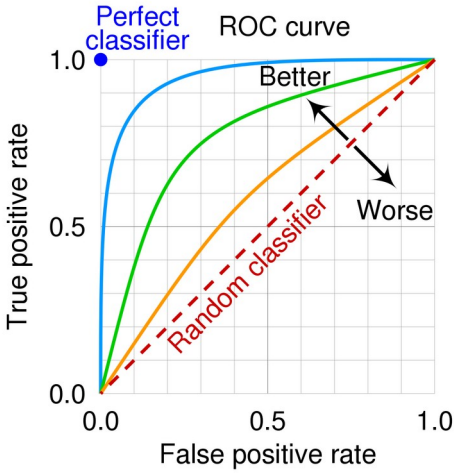


Figure 2. Interpretation of ROC Curve (Kılıç, 2023)

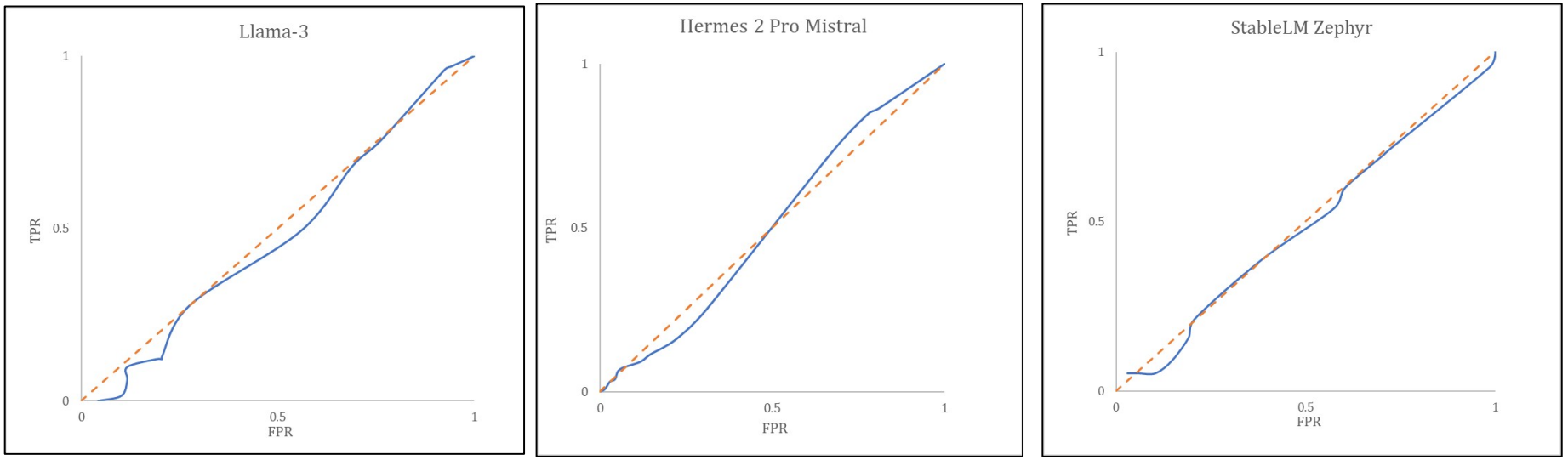


Figure 3. ROC curves for 3 LLM models (n = 200). AUC scores were 0.476, 0.503, and 0.486 (from left to right).

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Decision Tree Classification Model

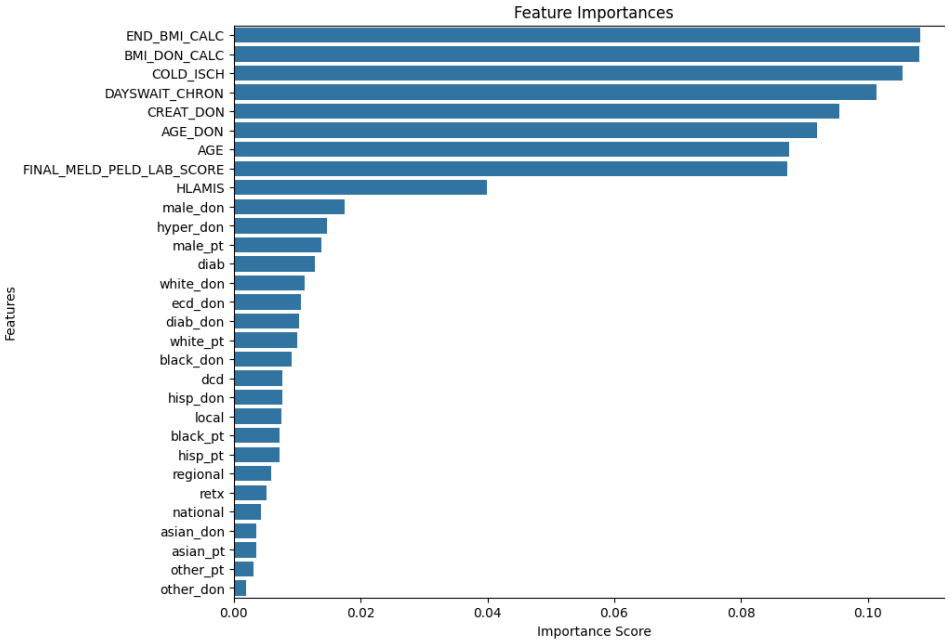


Figure 4. Feature Importances for 30 Patient and Donor Variables

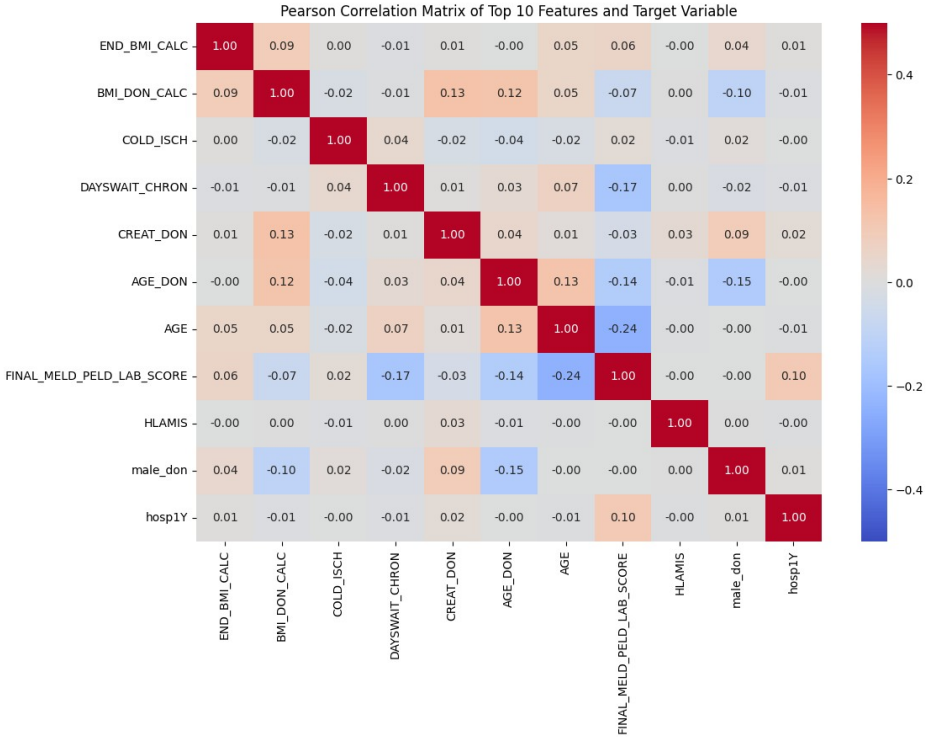


Figure 5. Pearson Correlation Matrix of Top 10 Feature and Target Variable

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Key Findings and Conclusion

- Binary classification was more accurate than probability classification
  - Best model: Hermes 2 Pro Mistral
  - Prompting bias
- Accuracy
  - High Recall
- Future improvements
  - More data
  - Optuna
  - Different models

Decision Tree Classification Model

-
Accuracy: 0.749656121045392
Precision: 0.7498277050310131
Recall: 0.9990817263544536

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Key Findings and Conclusion cont.

- ML>LLM in terms of accuracy
  - ML model identified key factors, consistent with previous research
  - Ex: increased BMI → liver complications
- LLM Limitations
  - Large computing power
  - Incorrect response output
- Our LLMs and prompts? Unsuitable for predicting patient outcomes
- Need further exploration of prompt-engineering for LLMs in medical contexts

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Individual Contributions

Janice:

- Created Binary Classification Code, experimented with variable combinations
- Presents Background, Excerpt prompting, and Binary Classification Results

Patrick:

- Created Probability Classification Code
- Performed hypothesis testing, produced ROC curve, AUC values
- Presents Methods, Probability Classification Results, and Key Findings and Conclusions cont.

Chris:

- Created Decision Tree Model
- Created Feature Importance Graph and Correlation Matrix
- Presents Study Objective, Decision Tree Classification Model, and Key Findings and Conclusion